

<Research Title>

Instructions

Proposals must be specific and detailed so the direction of the research is clear. If you're not sure about a given item, please mark it as such and we can discuss.

Add text using your assigned color for all tabs in this doc:

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Relevant Past Papers

How is it done today, and what are the limits of current practice?

- Things to include for each paper:
 - One sentence summary of the paper
 - Gap / limitation of the paper

Motivation

- Things to include:
 - What limitation or problem are you solving and how do you know it exists
 - Why is this limitation important
 - Why does your idea solve it
 - Why would your idea probably work

Key Ideas/Contributions/Novelty

- What are we doing that hasn't been done before? Things to consider:
 - What is the novelty of the idea with respect to the existing literature? What is "new"?
 - What is the main contribution (to science) of this paper? Are we creating an improved method or a benchmark? What new insights or findings are we providing? What research question are we answering that was not answered?

Methods

- How does your idea work? Describe the way you will get your results from the initial step. Make a diagram.

Experimental Setup

- How are you going to test your idea to prove that it works?
 - Think about social science experiments where researchers have a plan of what to make participants do, what to ask them and how to calculate results based on the responses.
 - What are your baselines for comparison? i.e control group
 - What models? What datasets? What metrics (if not accuracy)?
- What additional analysis do you plan on doing?
 - How are you going to prove that any improvements in the system are because of your idea alone, i.e. reduce confounding factors and do ablation testing
 - How does your idea impact the system? What metrics are you quantifying with?
 - Visualizations? Statistics?

Datasets and Evaluation

- Which datasets are you going to use to evaluate your method? Or, are you creating your own?
 - Reference relevant previous papers here.
- If you're training, what dataset will you use for that?
- What is the evaluation metric(s)?
 - Some tasks are straightforward to measure (e.g accuracy, for mathematical reasoning). Some are much harder (e.g LLM persuasive ability - think about how you would do this).

Benchmarks/Evaluation Sets

- Evaluation and comparison with existing systems.
 - What are your baselines?

Ideal Results

- What's the best case scenario/what do you hope to demonstrate?
 - What is your hypothesis? What results would prove it to be true?

Potential Limitations

- Computation limits, generalization limits, dataset limits, ethical limits?